324612(24)

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BE (6th Semester)

Examination, April-May, 2016

(Old Scheme)

Power System Protection and Switchgear

Time Allowed : 3 hours *Maximum Marks* : 80

Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of each question is compulsory. Attempt any **two** parts of (*b*), (*c*) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

Unit - I

1. (*a*) Define the terms : 2

(*i*) Backup relay

(*ii*) Operating or pickup level

(*b*) Explain IDMT relay. Why are IDMT relays widely used for overcurrent protection? 7

(*c*) What is Bucholz relay? Which equipment is protected by it? Discuss its working principle. For what types of faults it is employed? 7

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(*d*) Describe mho and offset mho relay. Explain its characteristics on RX diagram. 7

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Unit - II

2. (*a*) Enlist the different faults on the alternator. 2

(*b*) Explain the loss of excitation. How is it protected? 7

(*c*) Explain the percentage differential protection scheme of alternator. 7

(d) The neutral point of a 3-phase 20 MVA, 11 kV alternator is earthed through a resistance of 5Ω. The relay is set to operate when there is an out of balance current of 1.5A. The CTs have a ratio of 1000|5. What percentage of winding is protected against an earth fault and what should be minimum value of earthing resistance to protect 90% of the winding. 7

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Unit - III

3. (*a*) State the difficulties encountered in differential protection of transformers. 2

(*b*) A 3ϕ transformer having a line voltage ratio of 400V/33,000V is connected in star-delta. The CTs on the 400V side have a current ratio of 1000|5. What must be the ratio of CTs on the 33,000V side? 7

(*c*) Discuss the protection of ring main of parallel feeders. 7

(*d*) Explain carrier current protection scheme for long transmission line. 7

Unit - IV

4. (*a*) Define duality. 2

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(*b*) Discuss the rectifier bridge type phase comparator. 7

(*c*) Describe how mho relay is realised, using phase comparator. 7

(*d*) Explain the static impedance relay, using :
(*i*) An amplitude comparator
(*ii*) A phase comparator 7

Unit - V

5. (*a*) Define making current of circuit breaker. 2

(*b*) What is resistance switching? Deduce an expression for critical resistance. 7

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(*c*) Describe the construction and operating principle of application of vacuum circuit breaker. 7

(d) A 3ϕ circuit breaker is rated at 1250A, 2000 MVA, 33 kV, 4S. Find the rated symmetrical breaking current, making current and short time rating. 7

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